

CompTIA A+ Core 1 Playbook

IT Labs

Prepared by Garrick White

5.2.5 Connect Patch Panel Cables

You are a network technician for a small corporate network. Today, you moved a new workstation to Office 1, and now you need to connect the computer to the Ethernet local area network and the internet.

In this lab, your task is to connect the workstation to the wired network as follows:

In the Networking Closet:

- Use the appropriate twisted-pair cable to make a connection between the patch panel and the switch.
 - Use port Off 1 on the patch panel.
 - Use port 3 on the Cisco switch.

In Office 1:

- Connect the Office1 desktop computer to the local area network using the appropriate twisted-pair cable.
- Configure the Office1 desktop computer to obtain IP and DNS addresses automatically from the DHCP server on the local network.
- Use the Settings app to confirm that the workstation is properly connected to the local area network and the internet.

To configure DHCP:

1. Right-click the network icon at the bottom-right corner of the screen.
2. Select **Network and internet settings**.
3. Select **Ethernet**.

4. Scroll down to the IP/DNS assignment areas, then select **Edit**.

Explanation

Complete this lab as follows:

1. From the Networking Closet, connect the patch panel and switch.
 1. In the Inventory, expand **Cables**.
 2. Select a **Cat6a** cable.
 3. From the Selected Item pane:
 - Drag an **RJ45 Shielded Connector** to the Off 1 (Office 1) port on the patch panel.
 - Drag the other **RJ45 Shielded Connector** to port 3 on the Cisco switch (top row, third port from the left).
2. Connect the Office1 workstation to the local area network.
 1. From the top left, select **Floor 1 Overview**.
 2. In Office 1, select **Hardware**.
 3. Switch to the **Back** view of the computer.
 4. Under Shelf, expand **Cables**.
 5. Select **Cat6a Cable, RJ45**.
 6. From the Selected Component pane:
 - Drag an **RJ45 Shielded Connector** to the Ethernet port on the computer.
 - Drag the other **RJ45 Shielded Connector** to the open Ethernet port on the wall outlet.
3. Configure the workstation to obtain IP and DNS addresses automatically from the server on the network.
 1. Select the monitor to view the operating system.
 2. From the Windows taskbar, right-click **Start** and then select **Settings**.
 3. Select **Network & internet**.
 4. Select **Ethernet**.
 5. Next to IP assignment, select **Edit**.
 6. Change the Edit IP settings drop-down to **Automatic (DHCP)**, then click **Save**.
 7. At the top of the Settings dialog, select **Network & internet**.

The globe icon next to the word Connected indicates the machine is now connected to the network and can access the internet